BRAD GERVIN D. ESLABRA

Computer Engineering Graduate (GWA: 1.68) specializing in Embedded Systems Design

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SKILLS AND TECHNICAL PROFICIENCIES

Programming Languages: C, C++, Python, SQL, C# (.NET), Verilog / HDL
Domain Specializations: Embedded Systems Design, Embedded Software, Logic Circuits Design, PCB Design & Fabrication, Software Integration & Testing, Edge AI, Computer Vision, FPGA Programming
Hardware & Microcontrollers: ESP32, STM32, Arduino, ESP8266, Raspberry Pi
Developer Tools & Methodologies: Git (Version Control), Software Debugging & Validation, Technical Documentation, Object-Oriented Programming (OOP)

WORK EXPERIENCE

Research and Development Intern - Layad Circuits Electronics Engineering Supplies and Services **May 2025 - June 2025**

• Designed the schematic circuit and PCB layout for a custom ATmega-based microcontroller board and developed firmware to optimize peripheral control and performance.
•Designed and deployed an automated AT Command Tester application in C#, accelerating serial communication debugging and integration testing efficiency for ESP boards.
• Designed and tested a Flame and LDR Sensor Validation Board, streamlining component diagnostics and reducing manual hardware testing effort across cross-functional teams.
• Executed systematic code reviews, firmware verification, unit testing, and bug tracking, improving module reliability and upholding software quality standards prior to deployment.

OTHER PROJECTS

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curves Lane Safety Enhancement and Accident Detection

- Integrated 24GHz millimeter-wave radar sensors and RF communication modules across ESP32 boards, deploying the complete wireless sensor network on an actual rural blind curve in coordination with the local government.
- Trained a custom YOLOv8n computer vision model using Roboflow and Google Colab, successfully deploying and testing it on a Raspberry Pi 5 edge device.
- Developed an Avalonia UI monitoring dashboard utilizing Tailscale and MQTT protocols for real-time telemetry communication with the deployed Raspberry Pi 5 system.

Arduino Training Kit:

- Worked with a team to design a modular hardware learning kit equipped with fundamental sensors, modules, and documentation to streamline practical engineering education.
- Led the schematic design and pcb fabrication for the module.

Adjustable Linear Power Supply:

- Collaborated with a team to design and assemble a variable DC power supply unit based on the LM317T linear voltage regulator.
- Managed material sourcing, component selection, and the bill of materials (BOM).

EDUCATION

PHINMA University of Pangasinan **August 2022 - June 2026**

- Bachelor of Science in Computer Engineering

RELEVANT ACHIEVEMENTS AND LEADERSHIP

Quiz Bowl 5th Placer (CpE Regionals) **April 2026**
Logic Breadborading 4th Placer (CpE Regionals) **April 2025**
Member of ICpEP Region 1 **2023-2026**
Editor / Executive Unit Member of Student Council **June 2025 - June 2026**

- Received the Patriotism Award for exceptional leadership and civic dedication.
- Directed publication layouts, media workflows, and technical support for council operations.